

PROJECT PROPOSAL & DETAILED REPORT

CareGrid.io

A Unified Digital Platform for Hospital Care, Blood & Organ Coordination

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Department of Computer Science & Engineering

Submitted by — Team MedMatrix

Zafar Muhammad Amran — ID: 0112230946

Md Siam Rahman — ID: 0112331150

Jakaria Bhuiyan Shrabon — ID: 011221093

Madhobi Akter Munni — ID: 0112230481

Submitted to

Shayekh Ebne Mizan

United International University

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1. Introduction & Problem Statement

Hospitals today juggle several time-critical and record-heavy operations at once: matching organ donors with recipients, keeping enough blood on hand for emergencies, updating patients and their families, dispensing medicine safely, and settling bills at discharge. In most hospitals, especially in resource-constrained settings, these five functions are handled through separate registers, spreadsheets, phone calls, and disconnected software tools. This fragmentation creates delays exactly where delays are most dangerous — in emergency organ transplantation and urgent blood requirements — and it leaves plenty of room for human error in everyday record-keeping.

CareGrid.io is proposed as a single, secure, web-based platform that brings these five operations together: Organ Donation & Compatibility Matching, Smart Blood Bank & Emergency SOS, Daily Patient Update & Doctor Dashboard, Pharmacy & Prescription Management, and Billing, Insurance & Discharge. By unifying them under one system with a shared database and consistent access rules, CareGrid.io aims to cut emergency response time, reduce manual record-keeping mistakes, and give hospital staff, patients, and families a clear, shared view of what is happening at every stage of care.

1.1 Why This Problem Matters

- i. **Time-criticality:** Organ viability and emergency blood needs are measured in hours, sometimes minutes. A delay caused by a phone call not being answered or a paper register being out of date can cost a life.
- ii. **Fragmented systems:** Blood bank inventory, patient charts, pharmacy stock, and billing are frequently maintained in separate tools that do not talk to each other, causing duplicate data entry and inconsistent records.
- iii. **Family anxiety:** Without a reliable channel for updates, families of ICU or ward patients repeatedly call the nursing desk for information, adding pressure to already-stretched staff.
- iv. **Manual errors:** Hand-written prescriptions, manually tallied bills, and paper-based stock counts are all vulnerable to transcription mistakes that a digital, rule-based system can catch automatically.

2. Project Objectives

The primary goal of CareGrid.io is to design and (in later phases) implement a unified platform that achieves the following measurable objectives:

- i. Reduce the time taken to identify a medically compatible organ donor-recipient pair by automating compatibility scoring instead of relying on manual cross-checking.
- ii. Shorten emergency blood-shortage response time through automated SOS broadcasts to eligible, nearby donors.
- iii. Give doctors and nurses a single dashboard to log and visualize patient vitals, reducing time spent on manual charting.
- iv. Give patients' families a transparent, read-only view of daily updates, cutting down on repeated phone calls to hospital staff.
- v. Digitize the prescription-to-dispensing pipeline so that drug interactions and allergies are flagged automatically before harm occurs.
- vi. Automate billing calculations and discharge documentation so that patients leave with an accurate, itemized bill and a complete digital record of their stay.

- vii. Maintain strict data security and role-based access, since the system handles sensitive medical and financial information.

3. Project Scope

This document defines the intended scope of the Software Engineering Laboratory project. It covers five core modules, each addressing a distinct hospital function, unified by a common platform, authentication system, and database. The current deliverable is the project idea and feature specification; subsequent phases of the course will cover detailed system design (ER diagrams, use-case diagrams, architecture), implementation, and testing.

3.1 In Scope

- i. Organ donation compatibility matching and waiting-list management
- ii. Blood inventory tracking and emergency donor alerting
- iii. Patient vitals logging and a family-facing update portal
- iv. Ward and bed occupancy visualization
- v. Digital prescriptions with interaction/allergy checking
- vi. Pharmacy stock synchronization
- vii. Itemized billing, insurance claim tracking, and discharge summaries

3.2 Out of Scope (for this phase)

- i. Integration with external national organ-registry or insurance-provider APIs (proposed as future work)
- ii. Native mobile applications (the initial platform is web-based and responsive)
- iii. Telemedicine / video consultation features

4. System Overview

CareGrid.io is architected as a modular web platform. Each of the five modules is a self-contained functional unit with its own data models and workflows, but all modules share a common patient/user identity, a common hospital/ward context, and a common notification layer (used both by the Emergency SOS broadcast and by the family/attendant update portal). Role-based access control governs what each type of user — clinician, nurse, pharmacist, billing officer, blood-bank coordinator, patient, or family attendant — can see and do.

Aspect	Description
Team Name	MedMatrix
Project / Platform Name	CareGrid.io
Domain	Healthcare Technology — Hospital Management, Blood Bank & Organ Donation
Core Modules	5 (Organ Matching, Blood Bank, Patient/Doctor Dashboard, Pharmacy, Billing)

Aspect	Description
Primary Users	Doctors, nurses, pharmacists, billing staff, blood-bank coordinators, patients, family attendants

Table 4.1 — Project summary at a glance

5. Organ Donation & Compatibility Matching Engine

Organ matching is both time-sensitive and medically strict, so this module carries significant technical depth in its backend and database design. Its core purpose is to ensure that only medically compatible and sufficiently urgent matches are ever surfaced to clinicians, removing guesswork from a decision where mistakes are irreversible.

5.1 Compatibility Scoring Algorithm

The system automatically matches potential donors and recipients using four weighted factors: Blood Group compatibility, Tissue Typing (HLA compatibility), an Urgent Needs Index reflecting the recipient's medical urgency, and Geolocation/Distance between donor and recipient, which affects feasible transport time. Combining these into a single compatibility score lets clinicians see ranked matches instead of manually cross-checking donor and recipient files.

5.2 Recipient Waiting List Priority

Recipients are queued and prioritized using a clinically recognized urgency score — such as MELD (Model for End-Stage Liver Disease) or PELD (Pediatric End-Stage Liver Disease) — combined with the total time a recipient has already spent on the waiting list, so that both medical urgency and fairness are reflected in the queue order.

5.3 Organ Cold-Ischemia Time Tracker

Once an organ is retrieved, its viability window begins closing immediately. This feature provides live countdown timers for organs in transit — for example, roughly 4 hours for a heart and roughly 24 hours for a kidney — and raises escalating alerts as each organ's preservation threshold approaches, so logistics teams can react before an organ becomes unusable.

5.4 Anonymous Living Donor Registry

A secure donor portal allows living donors to register anonymously and complete eligibility screening checklists entirely online, lowering the barrier to registration while protecting donor privacy throughout the process.

6. Smart Blood Bank & Emergency SOS

This module digitizes blood inventory management end-to-end and introduces an automated emergency alert system designed to accelerate donor mobilization at the exact moment a critical shortage occurs — the scenario where minutes matter most.

6.1 Real-time Inventory & Expiration Alerts

Blood is not tracked as a single pooled quantity; the system monitors Whole Blood, Red Blood Cells, Platelets, and Plasma separately, since each component has its own demand pattern and shelf life. Automatic notifications fire when any component's stock drops below a safe threshold or when units approach their expiration date, preventing both shortages and wastage.

6.2 Emergency Broadcast System (SOS)

When an urgent ICU match is needed, the platform automatically sends SMS and email blasts to all registered donors of the required blood type, turning what is normally a slow phone-tree process into an instant, wide-reaching alert.

6.3 Donor Eligibility & Interval Tracker

To protect donor health, the system automatically blocks any donor who gave blood fewer than 56 days ago, enforcing standard medical donation-interval guidelines without relying on staff to remember or manually check each donor's history.

6.4 Geofenced Donor Search

An interactive, map-based interface displays nearby eligible donors positioned relative to the requesting hospital, helping coordinators identify the fastest realistic source of a match during an emergency.

7. Daily Patient Update & Doctor Dashboard

This module centralizes patient information into a dashboard that keeps clinical staff and patients' families informed together, reducing the communication burden that otherwise falls on ICU and ward staff throughout the day.

7.1 EHR Timeline & Vitals Logging

Daily temperature, blood pressure, oxygen levels, and medication charts are logged and visualized as line graphs, letting doctors and nurses spot trends at a glance instead of scanning rows of numbers on a paper chart.

7.2 Attendant / Family Portal

Family members are given read-only mobile and web access to the daily updates written by doctors or nurses, which removes the need for repeated phone calls to the ICU or ward desk simply to ask, 'How is the patient doing?'

7.3 Ward & Bed Management

A visual grid layout displays every hospital room and bed along with its current status — Occupied, Available, Cleaning, or Reserved — giving ward staff a fast, at-a-glance view for allocation decisions during admissions and transfers.

8. Pharmacy & Prescription Management (E-Prescriptions)

This module digitizes the full prescription-to-dispensing workflow, reducing manual transcription errors and improving patient safety through automated interaction and allergy checks that a busy human eye might miss.

8.1 Digital Prescription Builder

Doctors generate prescriptions digitally within the platform, and each prescription is automatically routed to the hospital pharmacy — eliminating illegible handwriting and lost paper slips as sources of dispensing error.

8.2 Drug Interaction / Allergy Warning

Before a prescription is finalized, the system flags dangerous interactions between the newly prescribed drug and the patient's currently active prescriptions or listed allergies, giving the prescribing doctor a real-time safety check.

8.3 Automated Stock Deduction

Whenever medicine is dispensed to a patient, the pharmacy's internal inventory is automatically reduced, keeping stock counts accurate in real time without a separate manual reconciliation step.

9. Billing, Insurance & Discharge Workflow

This module streamlines the financial and administrative closure of a patient's hospital stay, turning what is often a slow, error-prone, paper-heavy process into a fast, itemized, and fully digital discharge experience.

9.1 Itemized Billing Generator

The system automatically computes room charges (billed per day), pharmacy items dispensed, surgery fees, and doctor consultation fees, combining them into a single, transparent itemized bill for the patient.

9.2 Insurance Claim Status Tracker

Claims submitted to insurance providers are tracked within the platform with clear Approval or Rejection status updates, so billing staff and patients both know exactly where a claim stands.

9.3 Digital Discharge Summary

With one click, the system produces the patient's complete stay report — including lab test results and the post-discharge medication schedule — as a downloadable PDF, giving the patient a single authoritative document to take home.

10. Module Summary

The table below consolidates all five modules and their key capabilities for quick reference.

Module	Key Capabilities
Organ Donation & Compatibility Matching	Compatibility scoring (Blood Group, HLA, urgency, geolocation), waiting-list priority, cold-ischemia tracking, anonymous donor registry

Module	Key Capabilities
Smart Blood Bank & Emergency SOS	Real-time component-wise inventory, SOS broadcast alerts, 56-day eligibility enforcement, geofenced donor search
Patient Update & Doctor Dashboard	EHR vitals timeline, read-only family portal, visual ward/bed management grid
Pharmacy & Prescription Management	Digital prescription builder, drug interaction/allergy alerts, automated stock deduction
Billing, Insurance & Discharge	Itemized billing generator, insurance claim tracker, one-click digital discharge summary

Table 10.1 — Summary of all CareGrid.io modules and their key capabilities

11. Proposed System Architecture & Technology Direction

As this document represents the project idea and feature specification stage of the Software Engineering Laboratory course, the technical architecture below is proposed as a direction for the design phase rather than a final implementation decision. It reflects the kind of stack typically suited to a role-based, data-sensitive, real-time hospital platform of this shape.

11.1 High-Level Architecture

- i. A responsive web front-end serving distinct dashboards per user role (clinician, nurse, pharmacist, billing officer, blood-bank coordinator, patient/family).
- ii. A modular back-end exposing each of the five modules as a logically separate service or module behind a shared API layer, so modules can be developed and tested independently while sharing core identity and hospital-context data.
- iii. A relational database to preserve strong consistency for medical and financial records, given the correctness requirements around compatibility matching, prescriptions, and billing.
- iv. A notification/messaging layer (SMS + email) shared by the Emergency SOS broadcast and by family-portal update alerts.
- v. Role-based access control (RBAC) enforced at the API layer, since a single platform serves clinical staff, administrative staff, and patients/families with very different access needs.

11.2 Key Non-Functional Requirements

- i. **Security & Privacy:** Patient records, prescriptions, and billing data are sensitive; the system must enforce authentication, authorization, and encryption of data in transit and at rest.
- ii. **Reliability:** The Emergency SOS and organ cold-ischemia tracker are time-critical paths and must be resilient to partial system failures.
- iii. **Auditability:** Every compatibility match, prescription, and billing entry should be traceable to the staff member who created or approved it.
- iv. **Scalability:** The architecture should allow additional hospitals or wards to be onboarded without redesigning core modules.

12. Expected Outcomes & Impact

By unifying five interdependent healthcare functions into one platform, CareGrid.io is expected to deliver measurable improvements across three dimensions:

12.1 Faster Emergency Response

Automated compatibility scoring and SOS donor broadcasts remove manual lookup steps from two of the most time-critical processes in hospital care — organ matching and emergency blood requirements — directly shortening the time between a need being identified and a match being found.

12.2 Fewer Manual Errors

Automated stock deduction, drug-interaction checks, and itemized billing calculations replace error-prone manual steps across pharmacy and finance, reducing the everyday mistakes that accumulate in paper-based or disconnected systems.

12.3 Greater Transparency

The family portal, ward/bed visualization, and claim-status tracker give patients, families, and staff a shared, real-time view of what is happening — reducing uncertainty and the communication overhead it creates.

13. Conclusion

CareGrid.io brings together five interdependent healthcare functions — organ matching, blood bank coordination, patient monitoring, pharmacy management, and billing — into one integrated platform. By automating time-critical decisions such as organ compatibility scoring and emergency blood donor alerts, alongside day-to-day hospital operations like ward management and billing, the system is designed to reduce administrative overhead, minimize human error, and ultimately improve patient outcomes.

This document outlines the intended scope of the Software Engineering Laboratory project and will serve as the foundation for the subsequent design and development phases, including detailed system design, database schema, and implementation.

14. Team & Submission Details

Team Member	Student ID
Zafar Muhammad Amran	0112230946
Md Siam Rahman	0112331150
Jakaria Bhuiyan Shrabon	011221093
Madhobi Akter Munni	0112230481

Team Name: MedMatrix

Submitted to: Shayekh Ebne Mizan, United International University

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